ROCK, PAPER, SCISSORS

A Mini Project Presentation presented to the

College of Information Technology Education

PHINMA – University of Iloilo

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(ITE 260 – Computer Programming 1)

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Introduction

Rock, Paper, Scissors is a simple game played by two players who each choose one of three options: rock, paper, or scissors. In this mini project, the player competes against the computer using a Python program. The computer randomly selects one of the three choices, and the program compares the computer's choice with the player's choice. The result is displayed as a win, loss, or tie based on the rules of the game. This project demonstrates basic Python concepts such as input, output, lists, random selection, and conditional statements.

Objectives of the Study

General Objective

To develop a Rock, Paper, Scissors program using Python.

Specific Objectives

Specifically, this study aimed to develop a program that can:

1. Allow the player to enter rock, paper, or scissors.
2. Generate a random choice for the computer.
3. Compare both choices and display the correct game result.

Scope and Limitations

The program allows one player to play Rock, Paper, Scissors against the computer. It uses three choices—rock, paper, and scissors—and applies the standard rules to determine the winner. The program displays the computer's choice and announces whether the player wins, loses, or ties. The current version is text-based and runs in a Python environment. It does not include a graphical user interface, score tracking, multiple rounds, or a replay option. It also does not validate invalid player entries, so the player must enter one of the three listed choices.

Definition of Terms

Python: A programming language used to write and run the game.

Input: Information entered by the player through the keyboard.

Output: Information displayed by the program, such as the computer's choice and the result.

Random Selection: A process that lets the computer choose one option unpredictably from a list.

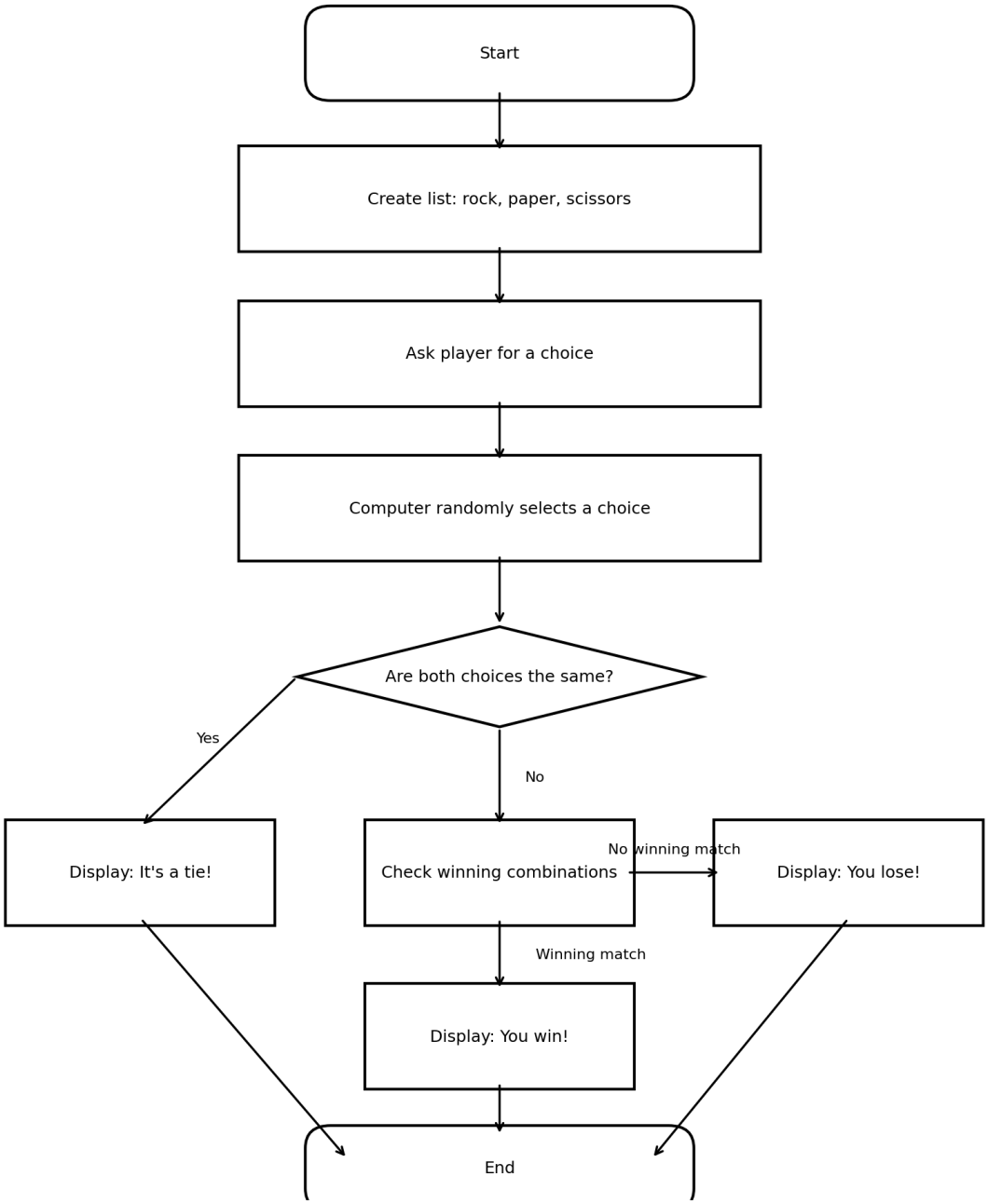
List: A Python data structure used to store the choices rock, paper, and scissors.

Conditional Statements: if, elif, and else statements used to decide which result should be displayed.

Tie: A result that occurs when the player and computer choose the same option.

Flow of the Data

The program starts by storing the three possible choices. It then receives the player's choice and randomly selects the computer's choice. The program compares the two choices and checks whether the result is a tie, a win, or a loss. Finally, it displays the result and ends.



Key Features

- Accepts a player's choice through keyboard input.
- Uses random.choice() to select the computer's move.
- Compares the player's choice with the computer's choice.
- Identifies a tie when both choices are the same.
- Uses conditional statements to determine whether the player wins or loses.
- Displays the computer's choice and the final result in the console.

Source Code

```
import random

choices = ["rock", "paper", "scissors"]

player = input("Choose rock, paper, or scissors: ").lower()
computer = random.choice(choices)

print("Computer chose:", computer)

if player == computer:
    print("It's a tie!")
elif (player == "rock" and computer == "scissors") or \
     (player == "paper" and computer == "rock") or \
     (player == "scissors" and computer == "paper"):
    print("You win!")
else:
    print("You lose!")
```